SYNTHETIC DEMONSTRATION DATA — NOT ACTUAL COLLEGE SAFEGUARDING RECORDS

SOUTH DEVON COLLEGE • SAFEGUARDING FORECAST AI

# Executive Briefing: Practical Staff Deployment & DSL Rota Engineering

Operational Planning Brief
Seed = 42 (Reproducible)

Multi-Model Count Benchmark (NB2 GLM vs. ZINB vs. Tweedie LightGBM) • UK KCSIE & GDPR Compliant Aggregate Forecasting

| WEEKLY DISPERSION RATIO | AUTUMN INDUCTION SURGE | EXAM & RESIT SURGE | 95% INTERVAL CALIBRATION |
|---|---|---|---|
| **24.10x** | **+57.9%** | **+45.7%** | **94.9%** |
| Mean 29.3/wk vs. Var 706.1 (High surge clustering) | NB2 IRR = 1.579x ($p < 0.0001$) in Weeks 1–6 | NB2 IRR = 1.457x ($p < 0.0001$) Nov & May/June | Empirical PICP (NB2 & ZINB vs. 95.0% target) |

## Five Practical Staff Deployment Pillars for College Leadership

Strategic & Operational Recommendations

### 1. Abandon "Average-Week" Flat Staffing (Core + Flex Deputy DSL Rota)

Dispersion Var/Mean = 24.1x

Across the academic year, college-wide referrals average **29.3 referrals/week** (~146/month). On paper, a core team of **4.5 FTE safeguarding officers** (at ~8 complex referrals/FTE/week = **36.0 referrals/week capacity**) appears sufficient. However, because weekly variance is **24.1x the mean**, demand fluctuates between **3–4 referrals/week** in placement/closure troughs and **55–135+ referrals/week** (with 97.5th percentile bounds exceeding 140) in peak weeks.

> **Deployment Action:** Operate a **Core + Flex Surge Rota**. Maintain core statutory DSLs year-round, supported by trained **Deputy DSLs (Curriculum Heads / Senior Pastoral Tutors)** whose teaching/admin remission is pre-scheduled to activate whenever the forecast 90th percentile bound ($Q_{0.90}$) exceeds 36 referrals/week.

### 2. Pre-Schedule Three Calendar Surge Windows (Autumn, Exam Crunch & Holiday Boundaries)

NB2 GLM Calendar IRRs

The Negative Binomial Governance GLM isolates three statistically significant calendar windows that drive peak DSL workload: **(a) Autumn Transition (Weeks 1–6, IRR = 1.58x, +57.9%)** driven by induction disclosures and secondary school handovers; **(b) Assessment & Resit Crunch (Nov Weeks 10–12 & May/June Weeks 36–41, IRR = 1.46x, +45.7%)** where 43% of demand stems from Mental Health & Wellbeing; and **(c) Pre-/Post-Holiday Boundary Weeks (IRR = 1.13x Pre-Holiday / 1.19x Post-Holiday)** when learners disclose home welfare risks before/after losing the college safe space.

> **Deployment Action:** (1) Ring-fence Deputy DSL hours in Weeks 1–3 for CPOMS/MyConcern secondary file handovers and Care Experienced/EHCP reviews. (2) Deploy **Wellbeing Advisers, Counselling practitioners, and Exam/ALS staff** as a "Tier 1 Triage Buffer" during Nov GCSE resits and May/June exams so statutory DSLs remain focused on child protection. (3) Ban discretionary DSL training/leave on the Thursday/Friday before holidays and Monday/Tuesday after return, and pre-book local authority MASH slots.

### 3. Weight Faculty Pastoral FTE by Cohort Vulnerability, Not Raw Student Headcount

16–18 Provision IRR = 3.64x

After controlling for active learner headcount via the exact log-exposure offset (`ln N_t`), **16–18 Study Programmes (EPYP)** represent 46% of learners (~3,450 students) but generate **61% of safeguarding referrals** (IRR = 3.64x, $p < 0.0001$). Granularly, *Foundation Learning & SEND* (8.4 referrals per 1,000/wk) and *Health, Social Care & Childcare* (5.6 per 1,000/wk) experience **2.5x–3.8x** the referral intensity of *Engineering, Marine & Automotive* (3.1) or *Adult Access* (2.2).

> **Deployment Action:** Replace flat per-capita pastoral staffing with **risk-weighted FTE allocation** tied to ILR statutory vulnerability (Care Experienced, EHCP, FSM) and IMD Deciles 1–3 concentration. Physically co-locate dedicated Link-DSLs inside **Foundation/SEND** and **Health & Social Care** hubs.

### 4. Link Student Finance (t–3) & Attendance Velocity (t–1) into a Monday Triage Huddle

1–3 Week Lead Time

Two behavioural lag indicators provide early warning before disclosures occur: **(a) 16–19 Bursary Emergency Hardship Claims at t–3 weeks** (`bursary_t3`, IRR = 1.047x, $p = 0.0397$) predict a referral rise three weeks later; and **(b) Lagged Attendance Drops & Tutor Pastoral Logs at t–1 week** are strongly collinear (VIF = 7.97 > 5.0), capturing the same acute disengagement wave one week prior to safeguarding escalation (`early_disengagement_index`, IRR = 1.043x).

> **Deployment Action:** Run a **15-minute Monday Morning Triage Huddle** connecting Student Finance (Bursary), Central Attendance, and the Lead DSL. A spike in emergency food/travel hardship claims in Week *t* gives the DSL team a **3-week lead time** to clear case backlogs and activate standby rota cover for Week *t+3*.

### 5. Reclaim Structural-Zero Weeks for Supervision & Maintain Summer Apprenticeship Duty

Low-Activity IRR = 0.094x (–90.6%)

Zero-Inflated Negative Binomial (ZINB) modeling confirms that reading weeks, work-placement blocks, and closures reduce on-campus referrals by **90.6%** (26%–40% zero-referral weeks at the curriculum level). However, **Apprenticeships** remain active in the workplace during July and August (0.72x summer demand multiplier vs. 0.22x for classroom cohorts).

> **Deployment Action:** Schedule mandatory KCSIE staff training, clinical supervision, multi-agency case audits, and DSL toil recovery during predictable low-activity weeks (Academic Weeks 8, 13, 21, 24, 28, 31–32), while assigning a dedicated **Work-Based Learning / Apprenticeship Safeguarding Duty Officer** across July–August.

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## Table 1: KCSIE / GDPR Explainability Audit — NB2 GLM Incidence Rate Ratios (IRR)

Offset = ln(Active Enrolments N_t)

| Predictor / Domain Feature | Coeff (β) | Std. Error | p-value | IRR = exp(β) | 95% CI | Operational & Staffing Interpretation |
|---|---|---|---|---|---|---|
| **Autumn Transition Surge** (`is_autumn_transition`) | +0.4568 | 0.0836 | <0.0001 | **1.5790x** | [1.34, 1.86] | **+57.9% surge** in Weeks 1–6 (induction & school file handover) |
| **Assessment & Resit Crunch** (`is_assessment_crunch`) | +0.3761 | 0.0753 | <0.0001 | **1.4565x** | [1.26, 1.69] | **+45.7% surge** during Nov GCSE resits & May/June exams/NEA |
| **Post-Holiday Return Week** (`is_post_holiday_week`) | +0.1707 | 0.0952 | 0.0731 | **1.1861x** | [0.98, 1.43] | **+18.6% increase** in the return week following college holidays |
| **Pre-Holiday Boundary Week** (`is_pre_holiday_week`) | +0.1262 | 0.0872 | 0.1478 | **1.1345x** | [0.96, 1.35] | **+13.5% increase** in the week immediately preceding breaks |
| **Emergency Hardship Claims t−3** (`bursary_t3`) | +0.0461 | 0.0224 | 0.0397 | **1.0472x** | [1.00, 1.09] | **+4.7% increase** per +1 bursary hardship claim/1,000 three weeks prior |
| **Composite Disengagement Index t−1** (`early_disengagement_index`) | +0.0422 | 0.0297 | 0.1555 | **1.0431x** | [0.98, 1.11] | **+4.3% increase** per +1 SD in lagged attendance drop + pastoral logs |
| **Prior Week Referral Rate t−1** (`referral_lag_t1`) | +0.0230 | 0.0058 | 0.0001 | **1.0232x** | [1.01, 1.03] | **+2.3% increase** per unit prior-week referral rate (peer contagion) |
| **16–18 Study Programme Share** (`provision_16_18_ratio`) | +1.2930 | 0.2070 | <0.0001 | **3.6436x** | [2.43, 5.47] | Higher 16–18 cohort share strongly elevates statutory referral demand |
| **Low-Activity / Placement / Closure** (`is_low_activity_week`) | −2.3695 | 0.1045 | <0.0001 | **0.0935x** | [0.08, 0.11] | **−90.6% reduction** during reading, placement, and closure weeks |

## Table 2: Expanding-Window Multi-Model Count Benchmark (Kaggle Evaluation Suite)

3 Expanding Origins • 26-Week Horizon

| Model Architecture | MAE | Surge MAE | NB Dev | RMSLE | RMSSE | WRMSSE | Pinball (0.90) | Pinball (0.975) | WSPL | 95% PICP |
|---|---|---|---|---|---|---|---|---|---|---|
| Legacy ML Baseline (Unconstrained L2 GBDT) | 9.079 | 14.907 | 2.3556 | 0.7131 | 0.4599 | 0.4718 | 3.0976 | 1.4146 | 0.1138 | 83.3% |
| **NB2 GLM (Governance Champion)** | **8.067** | **12.450** | **1.6116** | **0.5556** | **0.4641** | **0.5045** | **2.1205** | **0.7119** | **0.1038** | **94.9%** |
| **ZINB (Two-Part Sparse/Departmental Regime)** | **7.935** | **12.204** | **1.5937** | **0.5538** | **0.4515** | 0.4932 | **1.9654** | **0.6529** | **0.0980** | **94.9%** |
| Tweedie-Loss LightGBM (p=1.50, `init_score=ln N_t`) | 8.801 | 12.762 | 2.6572 | 0.7644 | 0.4538 | 0.4949 | 2.2863 | 0.9164 | 0.1116 | 80.8% |

## Table 3: Data Grain Dispersion & ZINB Regime

| Curriculum Area / Grain | Mean/Wk | Var/Mean | Zero % | ZINB ΔAIC |
|---|---|---|---|---|
| **College-Wide Weekly** | 29.29 | 24.10x | 10.3% | −3.96 (NB2) |
| **Health, Social Care & Childcare** | 7.20 | 9.67x | 34.0% | **+36.85 (ZINB)** |
| **Foundation Learning & SEND** | 6.14 | 8.29x | 34.6% | **+34.35 (ZINB)** |
| **Engineering, Marine & Auto** | 2.74 | 4.66x | 39.7% | **+13.32 (ZINB)** |
| **Construction & Building** | 5.52 | 5.59x | 27.6% | **+10.92 (ZINB)** |
| **Adult Access & Apprenticeships** | 3.15 | 4.49x | 36.5% | **+3.26 (ZINB)** |
| Digital, Computing & Creative | 4.54 | 6.65x | 35.9% | −1.64 (NB2) |

## Table 4: Sample DSL Weekly Triage Schedule

| Week Start | Expected (μ̂) | Q90 | Q97.5 | Staffing Status (Cap = 36/wk) |
|---|---|---|---|---|
| 2025-03-03 (W27) | 56.7 | 80 | 96 | HIGH ALERT: Mobilise Deputy DSLs |
| 2025-03-10 (W28) | 4.2 | 7 | 10 | WITHIN CAPACITY (Placement Wk) |
| 2025-03-17 (W29) | 40.6 | 59 | 71 | HIGH ALERT: Ring-Fence Triage |
| 2025-03-24 (W30) | 48.0 | 68 | 82 | HIGH ALERT: Pre-Easter Surge |
| 2025-03-31 (W31) | 4.1 | 7 | 9 | WITHIN CAPACITY (Easter Break) |
| 2025-04-07 (W32) | 2.8 | 5 | 7 | WITHIN CAPACITY (Easter Break) |
| 2025-04-14 (W33) | 36.4 | 52 | 63 | HIGH ALERT: Post-Easter Return |
| 2025-05-05 (W36) | 65.6 | 92 | 109 | HIGH ALERT: May/June Exam Crunch |